

The imperative paradigm: a decomposition of voice morphology across Formosan languages

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roadmap

1. introduction

2. imperative paradigm

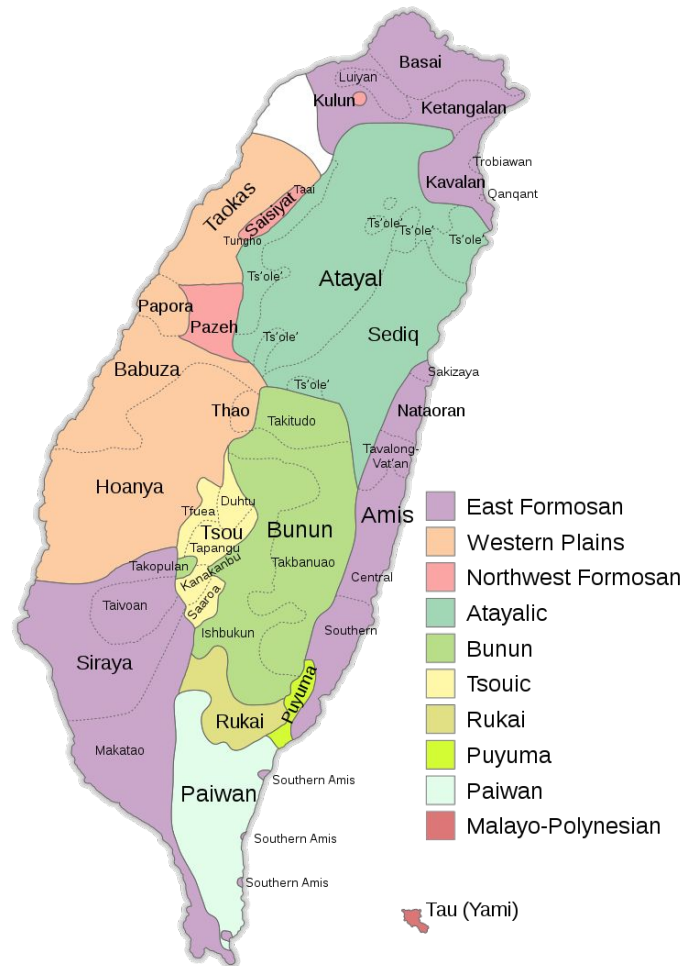
3. theoretical assumptions

- bundling
- irrealis and imperatives

4. proposed analysis

- projections: MT, E, Appl
- derivation

5. conclusion





main takeaway

Austronesian voice morphology—although seemingly in complementary distribution—should be **decomposed** into morphemes generated across **distinct syntactic projections**

common syntactic analyses:

set of morphemes w/ locus of spell-out delegated to a single head

(Rackowski 2002; V. Chen 2017; Erlewine et al. 2017, a.o.)

this talk:

an overlooked **morphological approach to voice systems...**

decomposition in Malagasy

(Guilfoyle et al. 1992; Pearson 2005; Travis 2010)

applicative bundling in Formosan

(J. Wu 2007, H. Chang 2009, B. Svensson 2023)



why decompose?

other motivations
to decomposing
voice morphology...

- **different orientation**—prefix vs. suffix:
morpheme-specific or structural difference?
- voices have **different syntactic distributions**,
cf. complementation, restructuring, adverbs

(V. Chen 2010, H. Chang 2017, Wurmbrand 2016, a.o.)

- **distinct historical origins** of voice markers:
*-en/-an as prepositions, *si- as light verb, etc.

(Starosta et al. 1981, Peterson 2017)

- **NEW!** imperative paradigm is **morphologically complex**

the imperative paradigm*

	INDICATIVE				IMPERATIVE			
	<u>AV</u>	<u>PV</u>	<u>LV</u>	<u>CV</u>	<u>AV</u>	<u>PV</u>	<u>LV</u>	<u>CV</u>
Atayal	⟨m⟩	-un	-an	si-	∅	-i	-i	-ani
Seediq	⟨m⟩	-un	-an	s-	∅	-i	-i	-ani
Paiwan	⟨em⟩	-in	-an	si-	-u	-u	-i	-an
Bunun	ma-	-un	-an	'is-	-a	-av	-av	'is- -av
Puyuma	⟨em⟩	-aw	-ay	-anay	∅	-u	-i	-ani
Tsou	⟨m⟩	-a	-i	-neni	-	-	-	-
Amis	⟨om⟩	-en	-an	sa-	∅	-en	-	sa- -en
Kavalan	⟨em⟩	-an	-	-	-ka	-ika	-	-

* voice forms simplified for exposition; see below sources of discussion of allomorphy and other complexities

source: 2018 CIP grammar series on Formosan languages, 2024 *Handbook on Formosan Languages* ed. Li, Zeitoun, & de Busse

the imperative paradigm*

	INDICATIVE				IMPERATIVE			
	<u>AV</u>	<u>PV</u>	<u>LV</u>	<u>CV</u>	<u>AV</u>	<u>PV</u>	<u>LV</u>	<u>CV</u>
Atayal	⟨m⟩	-un	-an	si-	∅	-i	-i	-an-i
Seediq	⟨m⟩	-un	-an	s-	∅	-i	-i	-an-i
Paiwan	⟨em⟩	-in	-an	si-	-u	-u	-i	-an
Bunun	ma-	-un	-an	'is-	-a	-a-v	-a-v	'is- -a-v
Puyuma	⟨em⟩	-a-u	-a-i	-an-a-i	∅	-u	-i	-an-i
Tsou	⟨m⟩	-a	-i	-nen-i	-	-	-	-
Amis	⟨om⟩	-en	-an	sa-	∅	-en	-	sa- -en
Kavalan	⟨em⟩	-an	-	-	-ka	-i-ka	-	-

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*if we take the imperative paradigm
as our starting point...*

...we would not immediately assume a
complementary nature of voice morphology!

- paradigm has **containment relation**
 - AV is least morphologically complex, then PV/LV, then CV
 - non-AV often share morphemes
- no overt “imperative” marker

	IMPERATIVE			
	<u>AV</u>	<u>PV</u>	<u>LV</u>	<u>CV</u>
Atayal	∅	-i	-i	-an-i
Seediq	∅	-i	-i	-an-i
Paiwan	-u	-u	-i	-an
Bunun	-a	-a-v	-a-v	'is- -a-v
Puyuma	∅	-u	-i	-an-i
Tsou	-	-	-	-
Amis	∅	-en	-	sa- -en
Kavalan	-ka	-i-ka	-	-

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theoretical assumptions

cross-linguistically, *functional projections can be bundled or fully articulated*, which impacts surface morphological outcome

“Voice-bundling”

(Pyllkänen 2008; Harley 2017)

“Split-IP”

(Thráinsson 1996; Bobaljik & Thráinsson 1998)

imperative constructions can be decomposed into **three main components...**

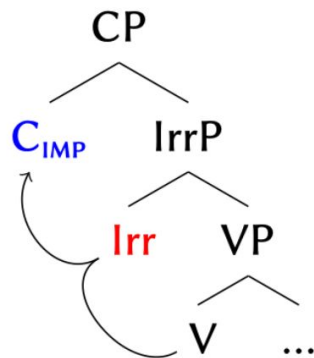
1) speech act C_{IMP} , 2) necessity modal, 3) irrealis Irr

work on subjunctive repair strategies in negative imperatives

(Han 1999, Rivero & Terzi 1995, Kaufmann 2012, Zanuttini, Pak & Portner 2012, a.o.)

positive imperative

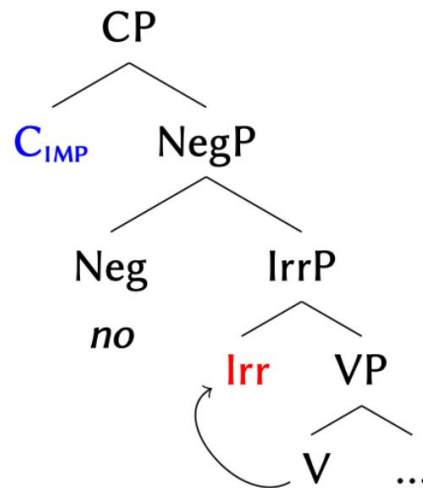
dá =me =lo
give.IMP.2SG =1SG =3SG
'Give it to me!'



V+Irr+C_{IMP} → “imperative”

negative imperative

no me lo des
NEG 1SG 3SG give.SUBJ.2SG
'Don't give it to me!'



V+Irr → “subjunctive”



proposed analysis

greater functional articulation of the extended Voice domain,
voice can be decomposed into three main components...

1) middle topic MT, 2) event closure E, 3) applicative Appl

hierarchy of projections

C > T > Neg > **lrr** > Asp > **MT** > **E** > Voice > **Appl** > v > √ROOT

INDICATIVE: YES! bundling → AV/PV/LV/CV is in *complementary distribution*

IMPERATIVE: NO! bundling → AV/PV/LV/CV is *morphologically decomposed*



independent motivations

previous work on Austronesian voice support each projection...

M(iddle)T(opic)

a low topic-like projection—movement to middle field:

pivots have topic-like qualities; pivots move to edge of Voice domain

(Aldridge 2004; Pearson 2005; Rackowski & Richards 2005; V. Chen 2017; Y. Wu 2024)

E(vent)

an event closure projection—used in non-finite marking:

default “AV” marking in reduced clauses; non-finite *-a marker in PAN

(Starosta et al. 1982; T.C. Chen 2010; Travis 2010; Wurmbrand 2016; Aldridge 2021)

Appl(icative)

an applicative projection—introduces oblique argument

(Rackowski 2000, Aldridge 2004; Pearson 2005; H. Chang 2009)



independent motivations

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M(iddle)T(opic)

a low topic-like projection—movement to middle field:

pivots have topic-like quality

(Aldridge 2004; Pearson 2005)

compromise between voice
as **argument vs. information**
structure alternations?

domain

(2017; Y. Wu 2024)

E(vent)

an event closure projection

default “AV” marking in red

(Starosta et al. 1982; T.C. Chen 2010; Travis 2010; von Stechow 2016; Aldridge 2021)

“voice morphology” bundles
of all of the above!!!

g:

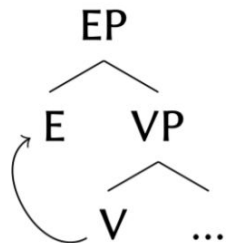
er in PAN

Appl(icative)

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actor voice

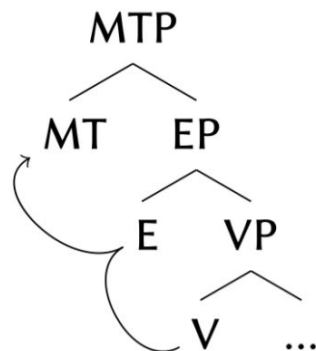


NOTE: MT doesn't project in AV constructions:

- AV vs. NAV binary distinction (Chang 2009)
- movement of agent is *vacuous* (Travis 1984)

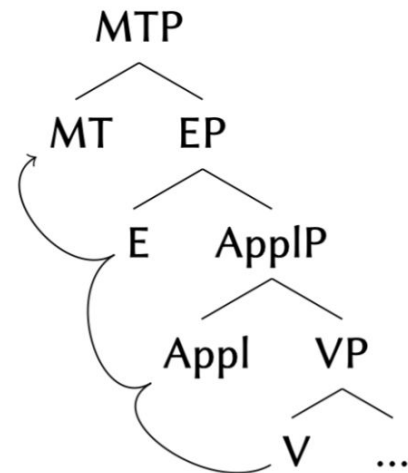
$E \rightarrow \text{"AV"}$

patient voice



$E+MT \rightarrow \text{"PV"}$

circumstantial voice



$Appl+E+MT \rightarrow \text{"cv"}$

INDICATIVE: YES! bundling $\rightarrow$ AV/PV/LV/CV is in *complementary distribution*

* setting LV aside for now, as it poses an interesting puzzle to be discussed later

roadmap

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Seediq

syncretism in **imperative, negative, negative imperative forms**

imperative

∅-ekan ido!
AV.IMP-eat rice
'Eat!'

puq-i ka ido!
eat-PV.IMP PIV rice
'Eat the rice!'

negative

ini=ku ∅-ekan ido
NEG=1SG.PIV AV.NEG-eat rice
'I didn't eat.'

ini=mu puq-i ka hlama
NEG=1SG.GEN eat-PV.NEG PIV candy
'I didn't eat the candy.'

negative imperative

iya ∅-ekan!
NEG.IMP AV.NEG.IMP-eat
'Don't eat!'

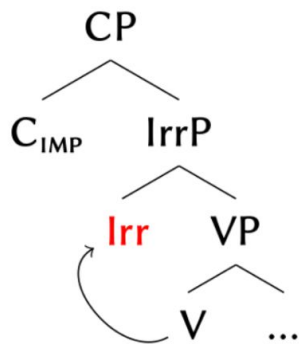
iya puq-i ka hlama!
NEG.IMP eat-PV.NEG.IMP PIV candy
'Don't eat the candy!'



irrealis morphology

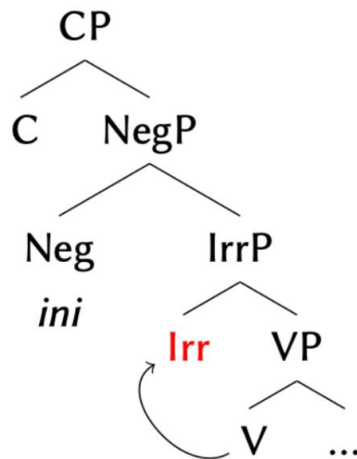
syncretism captured by presence of **irrealis projection IrrP**

imperative



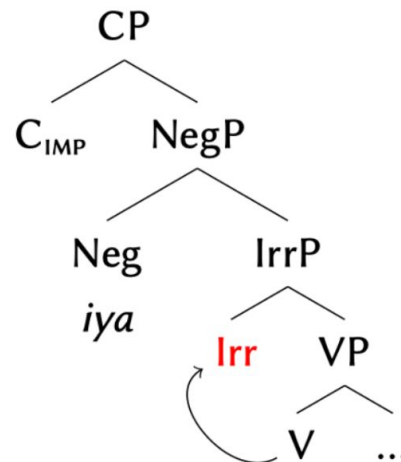
V+**Irr** → “irrealis”

negative



V+**Irr** → “irrealis”

negative imperative



V+**Irr** → “irrealis”



account

indicative vs. imperative paradigm easily captured through
contextual allomorphy on MT, E, Appl triggered by the IrrP projection

INDICATIVE: YES! bundling → AV/PV/LV/CV is in *complementary distribution*

IMPERATIVE: NO! bundling → AV/PV/LV/CV is *morphologically decomposed*

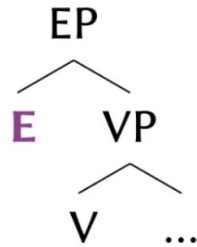
Seediq

	<u>AV</u>	<u>PV</u>	<u>LV</u> *	<u>CV</u>	Appl+E+MT ⇔ <i>s-</i>	Appl ⇔ <i>-an</i> / __ [IRR]
IND	⟨ <i>m</i> ⟩	<i>-un</i>	<i>-an</i>	<i>s-</i>	E+MT ⇔ <i>-un</i>	MT ⇔ <i>-i</i> / __ [IRR]
IMP	∅	<i>-i</i>	<i>-i</i>	<i>-an-i</i>	E ⇔ ⟨ <i>m</i> ⟩	E ⇔ ∅ / __ [IRR]

* setting LV aside for now, as it poses an interesting puzzle to be discussed later

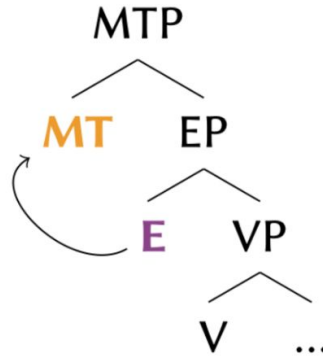
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actor voice



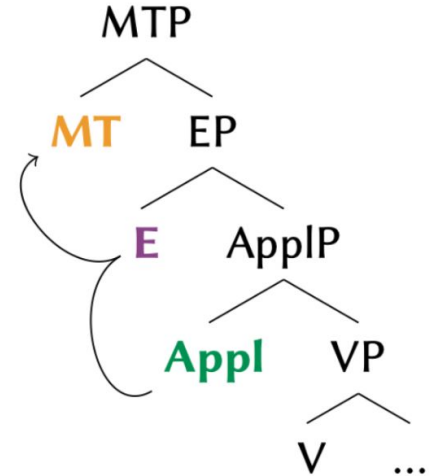
$E \rightarrow \langle m \rangle$

patient voice



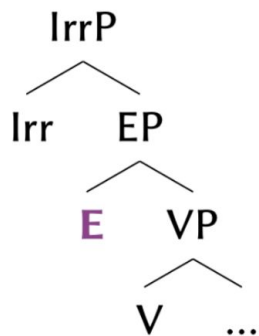
$E + MT \rightarrow -un$

circumstantial voice

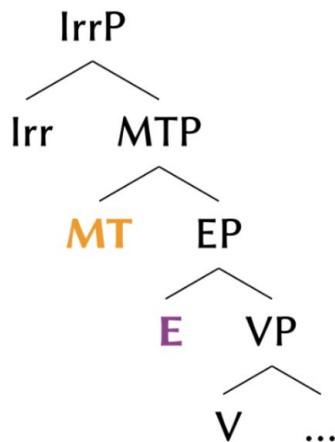


$Appl + E + MT \rightarrow s-$

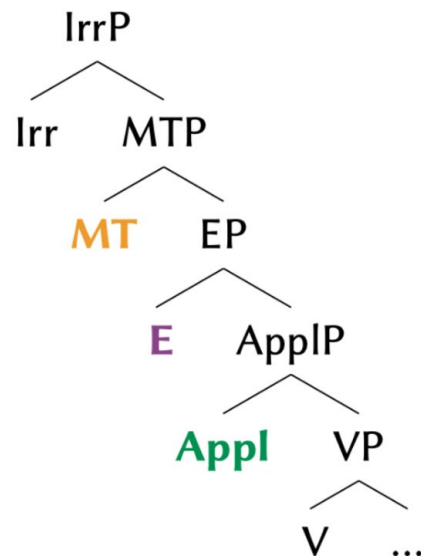
IMPERATIVE: NO! bundling → AV/PV/LV/CV is *morphologically decomposed*



E → ∅ _ [IRR]



E → ∅ _ [IRR]
MT → -i _ [IRR]



E → ∅ _ [IRR]
MT → -i _ [IRR]
Appl → -an _ [IRR]

the imperative paradigm*

	INDICATIVE				IMPERATIVE			
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Seediq	⟨m⟩	-un	-an	s-	∅	-i	-i	-an-i
Paiwan	⟨em⟩	-in	-an	si-	-u	-u	-i	-an
Bunun	ma-	-un	-an	'is-	-a	-a-v	-a-v	'is- -a-v
Puyuma	⟨em⟩	-a-u	-a-i	-an-a-i	∅	-u	-i	-an-i
Tsou	⟨m⟩	-a	-i	-neni	-	-	-	-
Amis	⟨om⟩	-en	-an	sa-	∅	-en	-	sa- -en
Kavalan	⟨em⟩	-an	-	-	-ka	-i-ka	-	-

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the imperative paradigm*

	INDICATIVE				IMPERATIVE			
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Atayal	<m>	-un	-an	-ci-	∅	-i	-i	-an-i
Seediq								
Paiwan		-m	-an	-er	-a	-a	-i	-an
Bunun	ma-	-un	-an	'is-	-a	-a-v	-a-v	'is- -a-v
Puyuma		-a-u	-a-i	-an-a-i	∅	-u	-i	-an-i
Tsou	<m>	-a	-i	-neni	-	-	-	-
Amis	<om>	-en	-an	sa-	∅	-en	-	sa- -en
Kavalan		-an	-	-	-ka	-i-ka	-	-

Puyuma shows **clear evidence for voice decomposition** in the CV.IND form w/ overt spell-out of all three heads in the expected order: Appl -an < E -a < MT -i

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other Formosan languages

Bunun

IND	ma-	-un	-an	'is-
IMP	-a	-a-v	-a-v	'is- -a-v

Appl+E+MT	⇔	'is-	Appl	⇔	'is- / __ [IRR]
E+MT	⇔	-un	MT	⇔	-v / __ [IRR]
E	⇔	ma-	E	⇔	-a / __ [IRR]

Paiwan

IND	⟨em⟩	-in	-an	si-
IMP	-u	-u	-i	-an

Appl+E+MT	⇔	si-	Appl+E	⇔	-an / __ [IRR]
E+MT	⇔	-in	MT	⇔	∅ / __ [IRR]
E	⇔	⟨em⟩	E	⇔	-u / __ [IRR]

Puyuma

IND	⟨em⟩	-a-u	-a-i	-an-a-i
IMP	∅	-u	-i	-an-i

Appl	⇔	-an	E	⇔	∅ / __ [IRR]
MT	⇔	-u/i	E	⇔	-a / __ [MT]
			E	⇔	⟨em⟩



locatives

- indicative forms are not syncretic
- different patterns of syncretism
 - PV and LV forms often identical
 - occasionally*, LV patterns with CV to the exclusion of PV
- finer distinctions in Appl projection
 - Appl**_{LOC} vs. **Appl**_{INST}
 - Puyuma: -u is MT, -i is Appl, -an is ?

	IMPERATIVE			
	<u>AV</u>	<u>PV</u>	<u>LV</u>	<u>CV</u>
Atayal	∅	-i	-i	-an-i
Seediq	∅	-i	-i	-an-i
Paiwan	-u	-u	-i	-an
Bunun	-a	-a-v	-a-v	'is- -a-v
Puyuma	∅	-u	-i	-an-i
Tsou	-	-	-	-
Amis	∅	-en	-	sa- -en
Kavalan	-ka	-i-ka	-	-

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summary

Austronesian voice morphology—although seemingly in complementary distribution—should be **decomposed** into morphemes generated across **distinct syntactic projections**

voice morphology

can be decomposed into three main components...

1) middle topic MT, 2) event closure E, 3) applicative Appl

INDICATIVE

MT, E, Appl are **bundled** → *forms portmanteaus*

IMPERATIVE

MT, E, Appl are **not bundled** → *spells-out separately*



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contributions

- greater articulation of the functional domain:
C > T > Neg > Irr > Asp > **MT** > **E** > Voice > **Appl** > v > √ROOT
- **bundling vs. decomposition**—in voice and imperatives
- Austronesian voice: **argument and information** structure
- **identifying basic building blocks** in “unusual” phenomena

future work:

- **CV**: most morphologically complex vs. elsewhere case
- **technical mechanics of bundling**: complex heads vs. feature checking, locus of spell-out, resolving orientation issue

thank you!

special thanks to...

SEEDIQ TEACHERS AND FRIENDS pai Habo, pai Lubi, Takun ptgesa, Pihu ptgesa, Mahung, Temu, Rabe, Pawan, Temi, Uma ptgesa, Tado ptgesa, Obing ptgesa

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